

# Product Recommendation for the Olist Marketplace: Capstone Final Report

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*(Written incrementally as each step landed. Sections for Steps 3 to 9 follow in order.)*

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## Step 1: Problem Understanding & Framing

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Olist is a Brazilian marketplace that connects small sellers to big storefronts. One number from the profiling run shaped the whole project: 96.9% of Olist's 96,096 customers have placed exactly one order, ever. That is a brutal retention picture. It also changes the business question. The question isn't "how do we rank products a bit better." It is whether a recommender can pull more customers into a second purchase at all, and whether it can widen what buyers see beyond the same bestsellers. There is a seller-side angle too. Olist sells small merchants on exposure, so a recommender that only ever surfaces popular stuff quietly breaks the marketplace's own value proposition.

This is a recommendation problem, but I split it into two stages on purpose. That is how production recommenders actually work, and each stage answers a different question. Stage one is candidate generation: given a customer, produce a shortlist of plausible products using item-item collaborative filtering, content similarity, matrix factorization, and a popularity baseline that I expect to be hard to beat on data this sparse. Stage two is a supervised ranker, a binary classifier scoring "will this (customer, product) pair convert as a delivered purchase" over engineered features. I considered simpler framings. Plain classification or customer clustering would have been easier to build, but neither produces a ranked product list, which is what a marketplace actually needs. The split has a practical payoff as well: the ranker is a feature-based model, and that is where explainability tools like SHAP genuinely work.

For candidate generation I track HitRate and Recall at  $K = 10, 50$  and  $100$ , plus  $NDCG@10$ , MRR and catalog coverage, with bootstrap confidence intervals on all of them. Hit rates over a 32,951-product catalog come out tiny, and I don't want to declare a winner on noise. The ranker gets ROC-AUC, PR-AUC and F1. One caveat travels with those three: they are computed against sampled negatives, so they compare rankers to each other and say nothing absolute. The decision-grade number is neither family on its own. It is the end-to-end table showing whether the full two-stage pipeline beats candidate generation on  $HitRate@10$  and  $NDCG@10$  for held-out purchases.

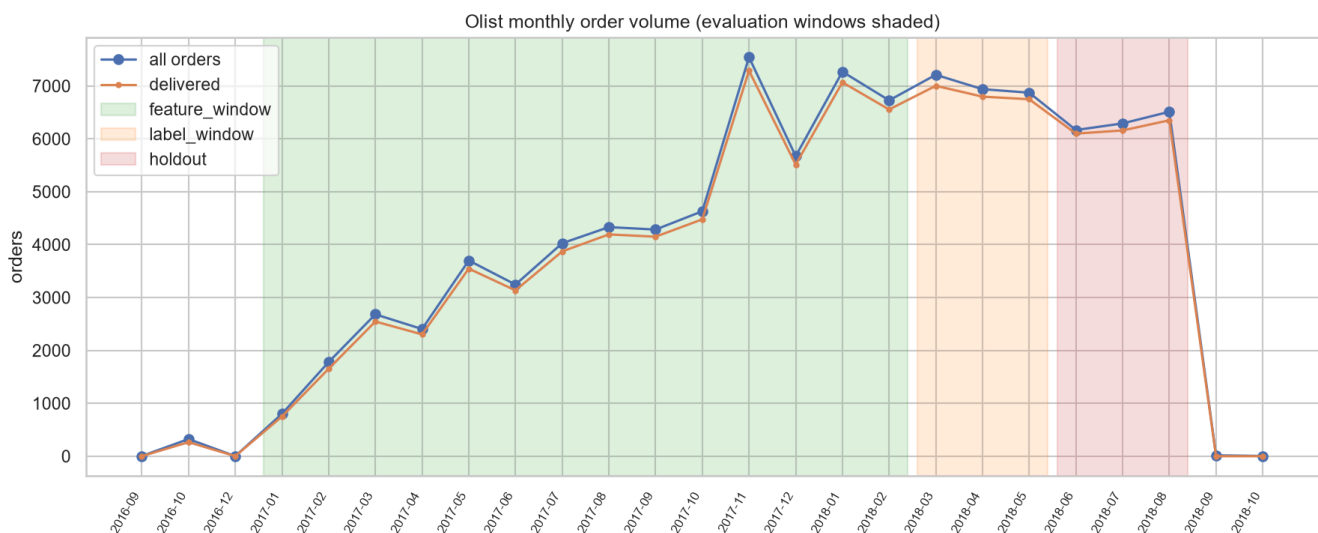
Each business KPI gets one number with a baseline from the profiling run. For the roughly 3% of customers with purchase history, the KPI is repeat-purchase rate, currently 3.12% (2,997 of 96,096

customers), proxied offline by the pipeline's HitRate@10 uplift over the popularity baseline on held-out repeat purchases. For the single-purchase majority the honest target isn't personalization, it is first-purchase cross-sell, which I proxy with category-level HitRate@10 on first orders against popularity. Seller-side exposure equity gets measured directly: the share of top-10 slots going to products outside the top sales decile. All three are offline proxies. Confirming real uplift would need an A/B test, and an offline capstone can't run one.

## Step 2: Data Collection & Understanding

I used the Brazilian E-Commerce Public Dataset by Olist ([kaggle.com/datasets/olistbr/brazilian-ecommerce](https://kaggle.com/datasets/olistbr/brazilian-ecommerce), CC BY-NC-SA 4.0): 99,441 real anonymized orders from 2016 to 2018 across nine relational tables covering orders, order items, customers, products, sellers, reviews, payments, geolocation, and a Portuguese-to-English category translation. I picked it over synthetic or ratings-style datasets because it is genuinely transactional. Real prices, freight costs, delivery dates and 1-to-5 review scores give the feature engineering something to chew on. Two trade-offs came with that choice: no customer demographics, which reshapes the fairness audit in Step 5, and extreme interaction sparsity.

The profiling notebook (`notebooks/01_data_overview.ipynb`) drove what I'd call the most important decision in the project: the evaluation window dates. Olist's monthly volume tells a clear story. A tiny pilot in late 2016 (329 orders, with November 2016 missing entirely), steady growth through 2017, a plateau around 6 to 7k orders a month in 2018, then a cliff after August 2018 where volume collapses to 16 and then 4 orders. Everything outside 2017-01 to 2018-08 is unusable. The three-window scheme is pinned to those dates in `configs/windows.yaml`: features from 2017-01 to 2018-02, ranker training labels from 2018-03 to 2018-05, holdout from 2018-06 to 2018-08. Every aggregate feature downstream is computed strictly inside the feature window.



The tables are in decent shape but not spotless, and a few quirks matter. `order_items` has one row per physical unit, so its 112,650 rows collapse to 101,987 unique customer-product pairs; anything building an interaction matrix has to dedupe first. The reviews table caught me out. `review_id` looks like a primary key but isn't: 789 values appear more than once (814 duplicate rows) and 547 orders carry more than one review. I keep only the latest review per order. Missingness reads as interpretable, not alarming: 88% of reviews have no title and 59% no message because most customers don't write text, and the 3% of orders missing a delivery date are mostly the undelivered ones. Prices are heavy-tailed. The median item costs about R\$75 while the p99 sits near R\$890 and the max at R\$6,735, which is why I winsorize price-derived features later rather than let a handful of luxury items stretch every scale. Review scores skew happy: 77.1% are 4 or 5. And 97.0% of orders reached `delivered` status, so filtering to delivered orders costs almost nothing. Geographically the dataset is lopsided, with 68.6% of customer records in the Southeast and just 1.9% in the North, which is what triggers the minimum-group-size rules in the fairness audit. A quick semantic pass over the columns (the notebook prints a per-table breakdown) shows mostly ids, categoricals and timestamps; the numeric signal sits in order items (price, freight) and product attributes (weight, dimensions), and free text exists only in review comments.



The full data dictionary lives in [docs/data\\_dictionary.md](#): every column across all nine files with types, missingness, units and allowed values. I'll be upfront about how it was made, since it doubles as my first

Step 9 deliverable. I had Claude draft it from schema profiles computed directly from the CSVs (`src/llm_docs.py`, with the prompt and raw output preserved in `docs/llm/`), then verified it by hand against the data. The draft was better than I expected, but the verification wasn't a formality. It confidently called `review_id` "unique," exactly the kind of plausible-sounding error you only catch by checking. That catch is now a documented correction in the dictionary, and the uniqueness check prints in the notebook so the number is reproducible.

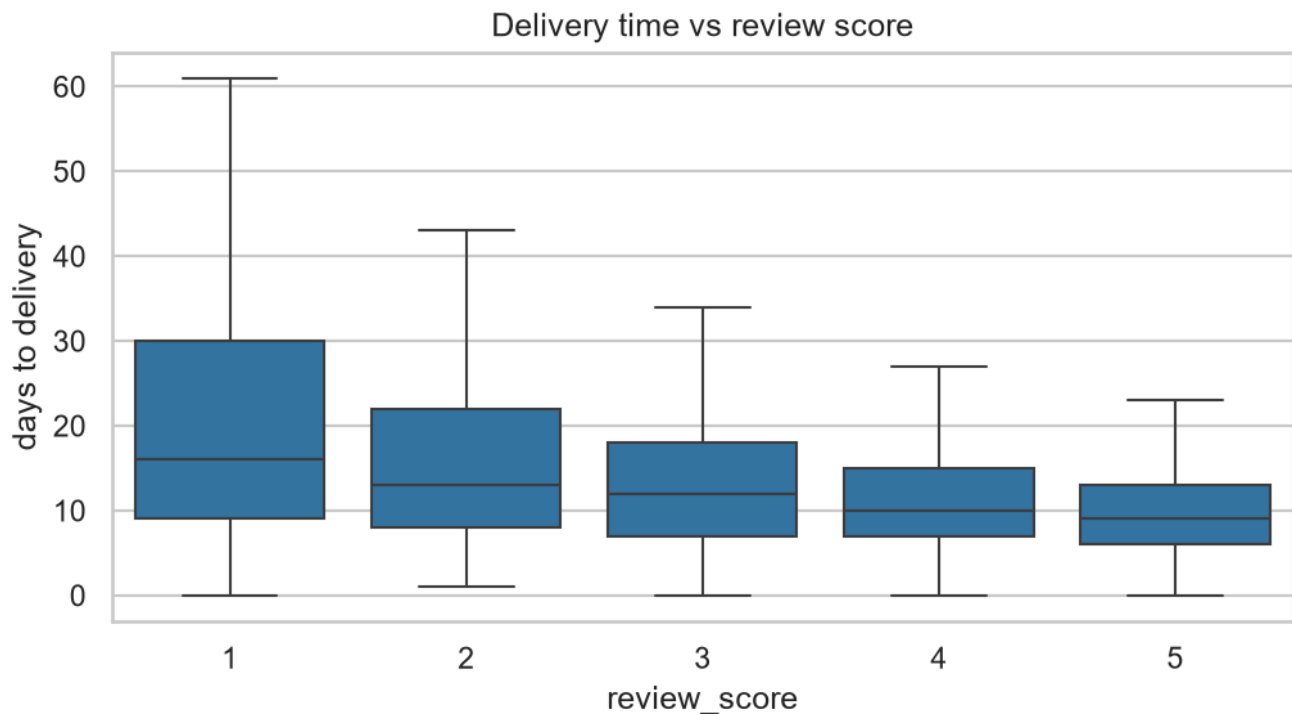
## Step 3: Data Preprocessing, Applied EDA & Feature Engineering

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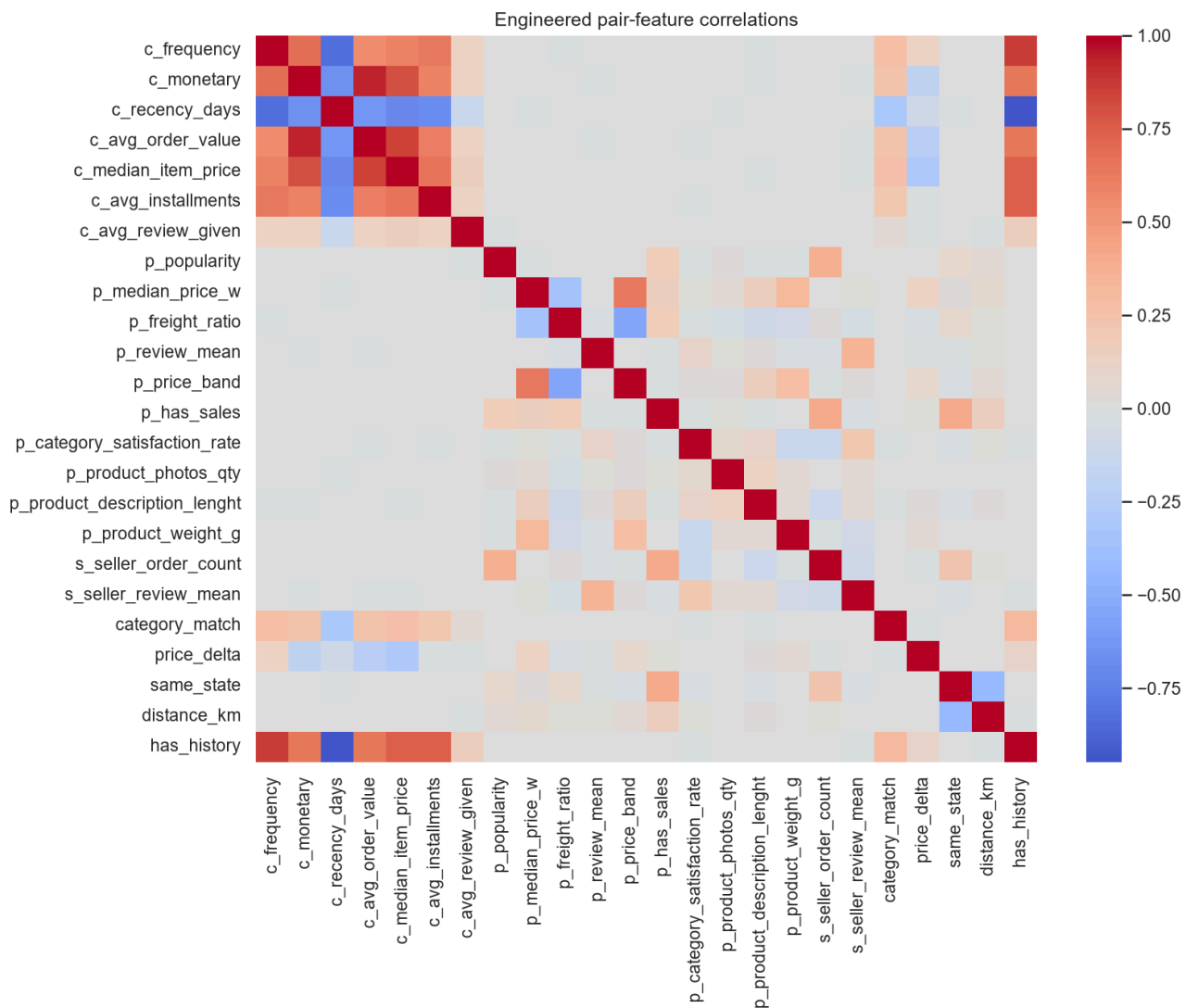
All the cleaning and feature logic lives in `src/features.py`, with `notebooks/02_eda_feature_engineering.ipynb` driving it, and every rule prints what it dropped or imputed. Filtering to delivered orders removes 2,963 of 99,441. The review dedupe (latest answer per order) removes 551 rows. Collapsing `order_items`' unit rows gives 102,425 order lines, and the product table needed 610 missing categories set to `unknown` plus 1,838 attribute values median-imputed. None of it silently changes the data, which was the point of returning the counts from every function. The window discipline from Step 2 is enforced here, and it turned out to need one rule I hadn't planned. A feature-window order can carry a review written after the cutoff. 3,805 reviews (6.7% of those joined to feature-window orders) were created past the cutoff, some well into the label window, so every review-based aggregate filters on `review_creation_date` and the notebook's audit cell now asserts both purchase and review-creation timestamps. Removing them nudged the average review given from 4.13 up to 4.17, because late reviews skew negative. I caught this in review after first shipping it wrong, which is a decent argument for auditing every timestamp column rather than just the obvious one.

The feature set covers three levels. Customers get RFM (recency, frequency, monetary), spend profile, average review given, installment habits and a preferred category. Products carry feature-window popularity, winsorized median price, freight-to-price ratio, review mean, a price band and the physical attributes, while sellers just carry volume and review reputation. Pair features are where the domain thinking sits: category match, price delta against the customer's median spend, a same-state flag, and a customer-to-seller haversine distance computed from the zip-prefix centroid lookup. Geography is *not* window-filtered, and that's deliberate. A customer's address is known at serving time, and window-filtering it would have thrown away the distance feature for the 98% of customers without purchase history. Cold-start fills are pinned to feature-window statistics (max recency 424 days, review mean 4.17, median customer-seller distance 453 km) so a pair's features don't depend on which batch it happens to be scored in. I also had to reinterpret one planned feature. The plan called for "category conversion rate," but Olist has no view or impression data, so conversion is unobservable; I substituted category satisfaction rate (share of the category's reviews at 4+), which is measurable and captures a similar quality signal.

The EDA backs the feature choices with numbers I found genuinely convincing. Median delivery time is 9 days on 5-star orders and 16 days on 1-star ones, and the 8% of reviewed orders that arrive late average 2.57 stars against 4.29 for on-time. That single contrast justifies every freight and distance feature in the set. Repeat buyers (2.92% of feature-window customers) spend more in total (R\$255 vs R\$136) but less per order, and lean harder on installments. Category demand isn't geographically uniform either; bed\_bath\_table's share swings 9.4 percentage points between regions within the feature window, which matters later because a popularity-only recommender would push the Southeast's taste onto everyone. Prices needed taming: the p99 winsorization cap lands at R\$1,102 and touches 205 products, about 1%.

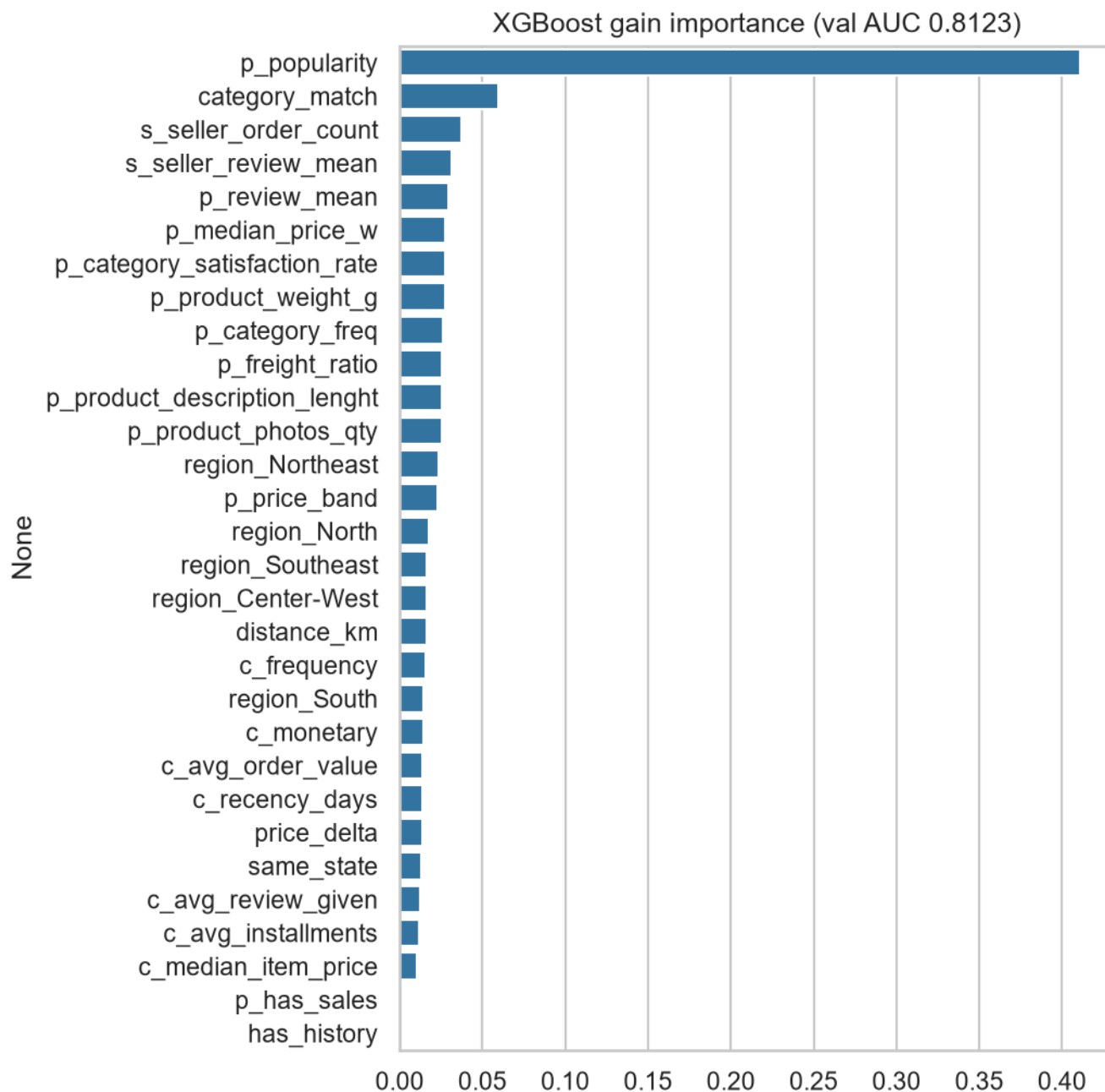


The correlation heatmap over the engineered features mostly shows what you'd expect, monetary and average order value at 0.94, spend measures moving together, but its most useful lesson is a warning. The strongest correlations in the matrix (recency by has\_history at -0.94, frequency by has\_history at 0.86) are structural artifacts of the cold-start fills, not behavior: when 98% of pairs share the same fill values, the fills correlate with the indicator by construction. I need to keep that in mind when reading feature importances and SHAP values later, since "importance" on a fill-dominated feature is really importance of being cold.



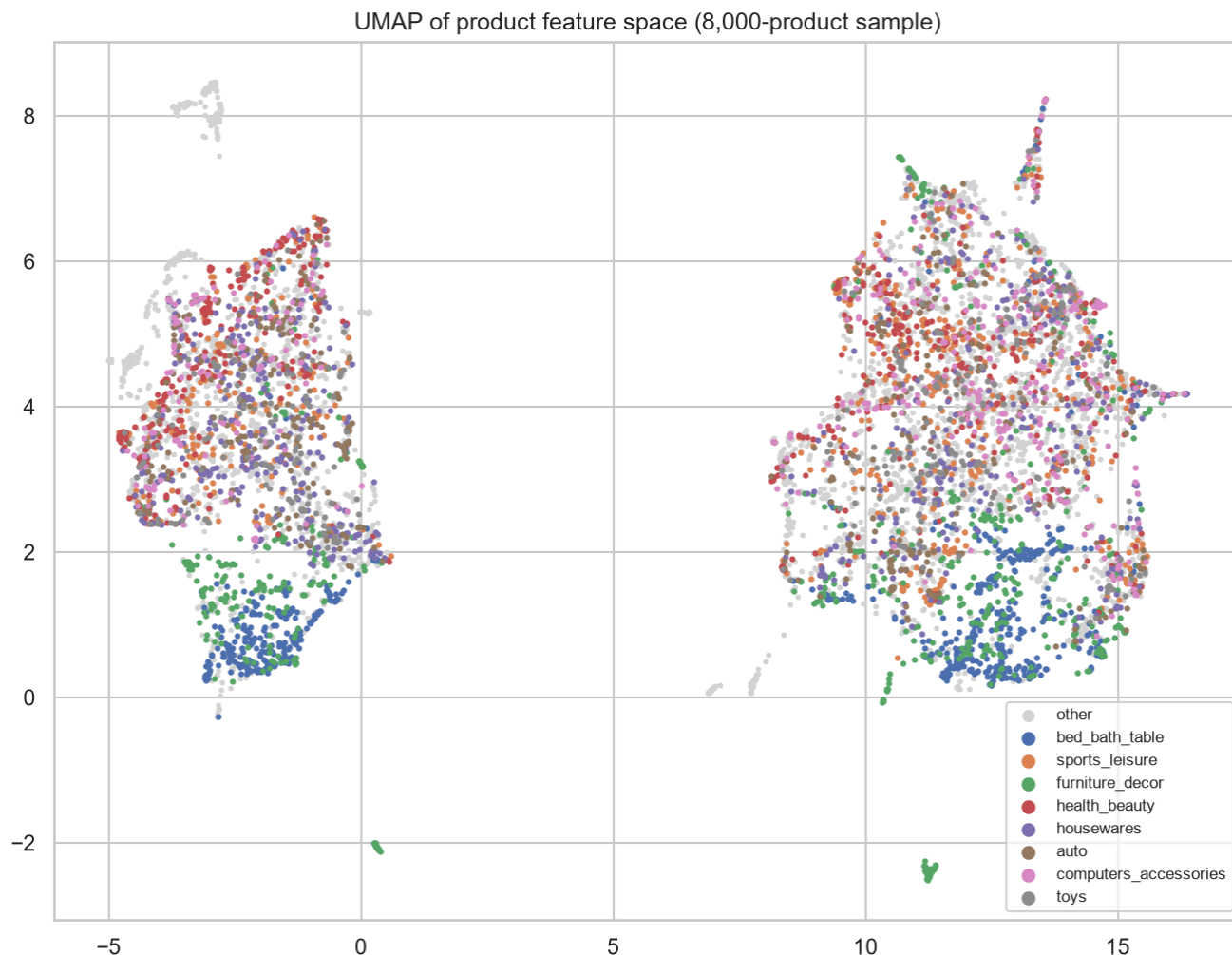
For the selection pass I assembled a preliminary ranking task: 21,260 label-window purchases as positives, uniform-random negatives at 1:4, splits grouped by customer. The cold-start number came out even starker than the Step 1 profiling suggested: only 2.09% of these pairs belong to a customer with feature-window history. The consequences show up immediately in the embedded selection. The L1 path zeroes 13 of 30 features at  $C=0.01$ , XGBoost's gain ranking puts product popularity at 0.41 with nothing else above 0.06, and the drop set is dominated by the six customer behavioral features that are cold for 98% of pairs. Picking  $C$  deserves a note, because the obvious rule gives a strange answer:  $C=0.001$  actually has the best validation AUC (0.7334), but it collapses the model to four popularity-dominated features, stranding the domain features I expect to matter once hard negatives exist. I went with  $C=0.01$  as the sparsest setting that keeps every feature family represented, and since  $C$  was chosen on the same validation set as the check below, I treat this whole pass as a screen. Retraining after the drop moved logistic regression from 0.7240 to 0.7244 AUC and XGBoost from 0.8123 to 0.8146, so the removed features were dead weight at best. One caveat: with random negatives this task is popularity-easy, so the

selection stays provisional until notebook 03 re-runs it with hard negatives from the candidate generators.



Dimensionality reduction rounds out the step. PCA on the 13 standardized product features needs 10 components to reach 90% variance, so the feature set isn't carrying much redundancy. That is good news for the trees, and a sign that dropping components would actually cost information. The UMAP projection of the same space (8,000-product sample) still shows visible category clustering, which is exactly the structure the content-based candidate generator will lean on. For encoding, I one-hot

encoded the five regions, frequency-encoded the 71 categories, and put a StandardScaler inside the linear pipelines only, since tree splits don't care about monotone rescaling.



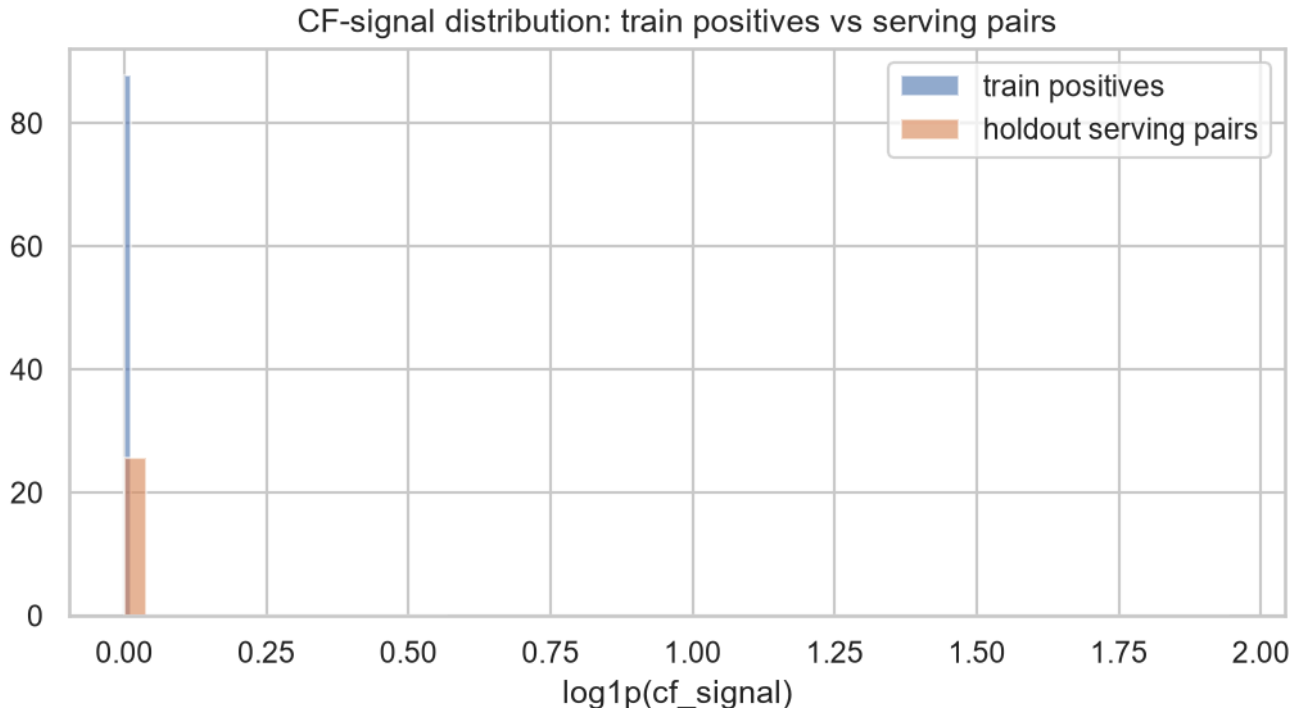
## Step 4: Model Implementation & Comparison

Five candidate generators went head to head under the leave-last-order-out protocol: each of the 2,789 repeat buyers' last orders held out, Stage-1 artifacts fitted on the remaining 96,883 usable-range purchases, 840 users reserved for tuning the SVD rank and hybrid weight, and the untouched 1,949 producing the comparison table. Since hit rates over a 32,951-product catalog are tiny numbers, every headline metric carries a bootstrapped 95% CI, and where two models were scored on the same users I also computed the paired per-user difference, which is far more powerful than eyeballing two overlapping intervals. The hybrid at  $w=0.25$  tops  $\text{HitRate@10}$  with 0.203 [0.184, 0.221] against content-based at 0.171, and the paired difference of +0.032 [+0.022, +0.042] makes that a decisive win, not a graze. It is also ahead deeper in the list (hit@50 0.241 vs 0.209, hit@100 0.260 vs 0.231, NDCG@10 0.151 vs 0.141) with comparable coverage (0.371 vs 0.377). Item-item CF (0.055), SVD



(0.054) and popularity (0.031) trail far behind. Worth admitting about the protocol itself: leave-one-out trades temporal strictness for evaluation-set size, since artifacts see purchases that postdate some users' held-out orders. The temporally clean numbers are the end-to-end table below, which is the decision table anyway. And the hybrid weight is a coarse choice, not a tuned claim; the validation spread across  $w$  was 0.0036 on 840 users, well within noise.

Repurchase drives the table. 69% of content's hits and 62% of the hybrid's are products the customer had already bought, which the seen-item policy allows on purpose and the table reports in its own column. For repeat buyers, "you'll buy again what you bought before" is the single strongest signal in this dataset, and it is also why popularity wasn't the unbeatable baseline the recommender literature warns about: popularity can't exploit that channel. On the category-level view (a 71-way space with workable base rates), CF actually leads at 0.55; it is better at guessing the *kind* of thing you'll buy than the exact product. SVD improves with rank but stays far below content and hybrid at every rank tried, topping out at 0.054 with  $k=256$ , itself at the edge of the swept grid. I didn't chase larger ranks for a model that isn't a contender. Olist's basket structure (1.04 distinct products per delivered order) leaves item-item CF with only 8,710 nonzero similarity pairs, which also kills the ranker's co-purchase feature: `cf_signal` is positive for 0.1% of training positives, and the sanity plot shows that outright. The feature I designed as the creativity hook is, on this dataset, nearly dead. I kept it and flagged it; the models mostly route around it anyway.



The ranker task uses the pinned protocol: label-window purchases as training positives, holdout-window purchases for evaluation, four negatives per positive, half from the customer's own Stage-1

candidate list and half uniform random. All three models were tuned with GroupKFold by customer from the grids in `configs/models.yaml`, and every run went to MLflow. XGBoost and random forest are effectively tied on the holdout (AUC 0.864 vs 0.865, PR-AUC 0.614 vs 0.619), both clearly ahead of logistic regression at 0.785. The random forest pays for its fit with a 0.945 training AUC, an overfitting gap I return to in Step 5. Classification thresholds come from out-of-fold predictions rather than training-fit probabilities, because the trees' training probabilities are optimistically separated and would bias their thresholds; with the honest thresholds, F1 lands at 0.578 for XGBoost, 0.569 for RF and 0.475 for LR. The standing caveat applies: these numbers compare rankers against each other under one sampling scheme, not in absolute terms.

The end-to-end table is where the pipeline decision actually gets made, because it is the only place candidate generation and re-ranking meet on one metric. For all 18,390 holdout customers: route (98.5% cold to regional popularity, 1.5% warm to hybrid top-50), re-rank with each model, keep ten, score against what they really bought. Re-ranking with XGBoost gives the best overall HitRate@10 at 0.0146 vs 0.0122 for Stage-1 order alone, and the paired per-customer difference of +0.0024 [+0.0008, +0.0041] means the +20% relative lift is small but real. Logistic regression re-ranking actively destroys value (0.0061), a concrete demonstration that a weak ranker is worse than no ranker. The most interesting wrinkle is the warm cohort. For the 277 customers with history, the raw hybrid ordering looks better than XGBoost re-ranking (0.0578 [0.033, 0.087] vs 0.0325 [0.014, 0.054]), but those intervals overlap heavily; at roughly 16 hits vs 9, the comparison is directional and underpowered. The tempting move is to re-rank only the cold route, but choosing that after seeing this table would be fitting to my own evaluation, so it goes in the future-work list with a requirement of fresh validation. The final pipeline is hybrid candidates plus XGBoost re-ranking, with the warm-cohort caveat up front.

For reproducibility: seeds are fixed at 42 throughout, the chosen configuration is written to `models/chosen_config.yaml`, all metrics land in `models/metrics.json`, artifacts (similarity matrix, content matrix, feature-window SVD factors, popularity vector, the three rankers) are saved under `models/` (the random forest squeaked under the repo size limit at 53 MB compressed), and every model run is logged to MLflow's local store. I considered a neural recommender (NCF-style) and decided against it. With 1.04 products per order and a repurchase-dominated signal, the data can't feed a model whose whole advantage is learning interaction structure; the SVD result is the empirical version of that argument, and I'd defend that reasoning over any reflexive "deep learning is better."

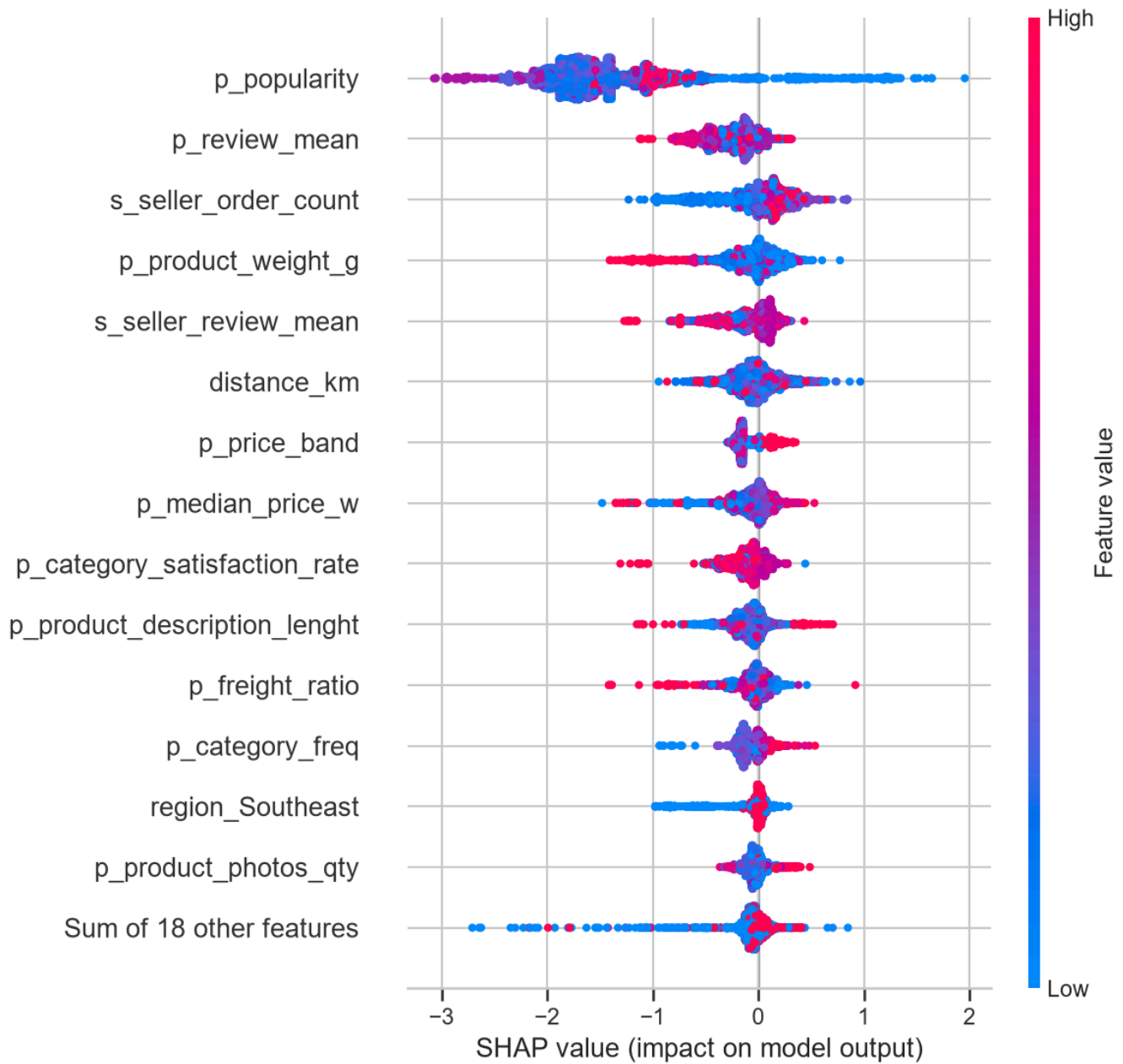
## Step 5: Critical Thinking, Ethical AI & Bias Auditing

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Everything in this step runs on the population that actually matters: each holdout customer's Stage-1 candidate list, 919,500 pairs from 18,390 customers, scored by the chosen XGBoost ranker with the decision rule  $\hat{y} = 1$  iff  $\text{score} \geq \tau = 0.272$ , the threshold fixed out-of-fold back in Step 4. I didn't audit on per-user random negatives, because a fixed sampling ratio forces every region's base rate to that ratio,

which makes demographic parity trivially and meaninglessly "fair." On the real candidate lists, regional base rates differ (0.0005 in the South to 0.0009 in the Northeast), and every fairness table below prints them alongside its metrics. One conditioning fact matters up front: a purchase that never made a candidate list is invisible to this audit, so the notebook also prints per-region Stage-1 retrieval recall, and it isn't uniform, 0.043 in the Northeast against 0.025 to 0.032 everywhere else.

The explainability stack starts global. SHAP's beeswarm puts product popularity far ahead of everything else (mean  $|\text{SHAP}|$  1.49 against 0.26 for product review mean), with seller volume, product weight and distance filling out the top ten. The direction is the interesting part. Within a candidate list, high popularity pushes scores *down*, because the hard negatives were themselves drawn from popular candidate lists, so the model learned "popular but not purchased" as its dominant negative pattern. The PDP makes the same point graphically: partial dependence peaks sharply for products in the popularity-20-to-60 band and craters above ~75, so the model actively inflates mid-popularity items. That pair of observations ends up explaining the mitigation failure below, which is the best argument I can make for doing explainability before intervening. The local views agree. The warm customer's top candidate scores 0.864 on the strength of purchase-history features; the cold customer's best candidate reaches 0.327 carried by product and seller attributes (photos +0.45, seller volume +0.42, seller reviews +0.28, even the 856 km distance +0.20) while its high popularity subtracts 0.58 and the cold-fill features net almost exactly zero. LIME on the same two cases matches SHAP on the popularity direction and the seller-volume contribution, and locally surfaces `cf_signal` and category match as well; agreement between two methods with entirely different mechanics is about as much comfort as local explanation offers. The Step 3 warning still applies throughout: importance on a cold-fill-dominated feature is partly importance of being cold.



On limitations, the three the rubric names each have a concrete face here. Imbalance: the ranker's training task is 20% positive by construction, and every AUC/PR-AUC/F1 I report is relative to that sampling design, a caveat that has followed the numbers since Step 4. Leakage: the three-window protocol is the defense, and it wasn't hypothetical, the review-creation leak I caught and fixed in Step 3 (3,805 reviews on feature-window orders, 6.7% of those joined, created past the cutoff) is exactly the failure mode the audit cells now assert against. Overfitting: the random forest's train AUC of 0.945 against 0.865 on holdout is the cautionary exhibit, XGBoost's 0.925 gap is smaller, and what discipline exists comes from depth 6, subsampled columns and the tuning grids. Beyond the named three sits the popularity feedback loop: a deployed recommender serving 98.5% of customers from popularity lists will reinforce the very concentration the exposure audit measures.

Now the audit itself. Olist collects no gender, race, or age, so I take each in turn rather than waving at "no demographics." There is no gender field, and inferring it from first names, a common hack, would inject its own error plus an ethical problem worse than the one it solves. Race isn't collected either and isn't responsibly inferable in the Brazilian context; age is simply absent. What the data does carry is geography, and in Brazil geography is a defensible socioeconomic proxy. By IBGE's PNAD Contínua (2018), mean household income per capita runs from about R\$815/month in the Northeast and R\$886 in the North to R\$1,502 to R\$1,554 in the Center-West, Southeast and South, roughly a factor of 1.9 between the poorest and richest regions, and my customer base skews hard toward the rich end (68.6% Southeast, 1.9% North). The proxy's own weakness gets named too: region-level income says nothing about any individual customer (the ecological fallacy), so this audit speaks about group-level service quality, not individuals.

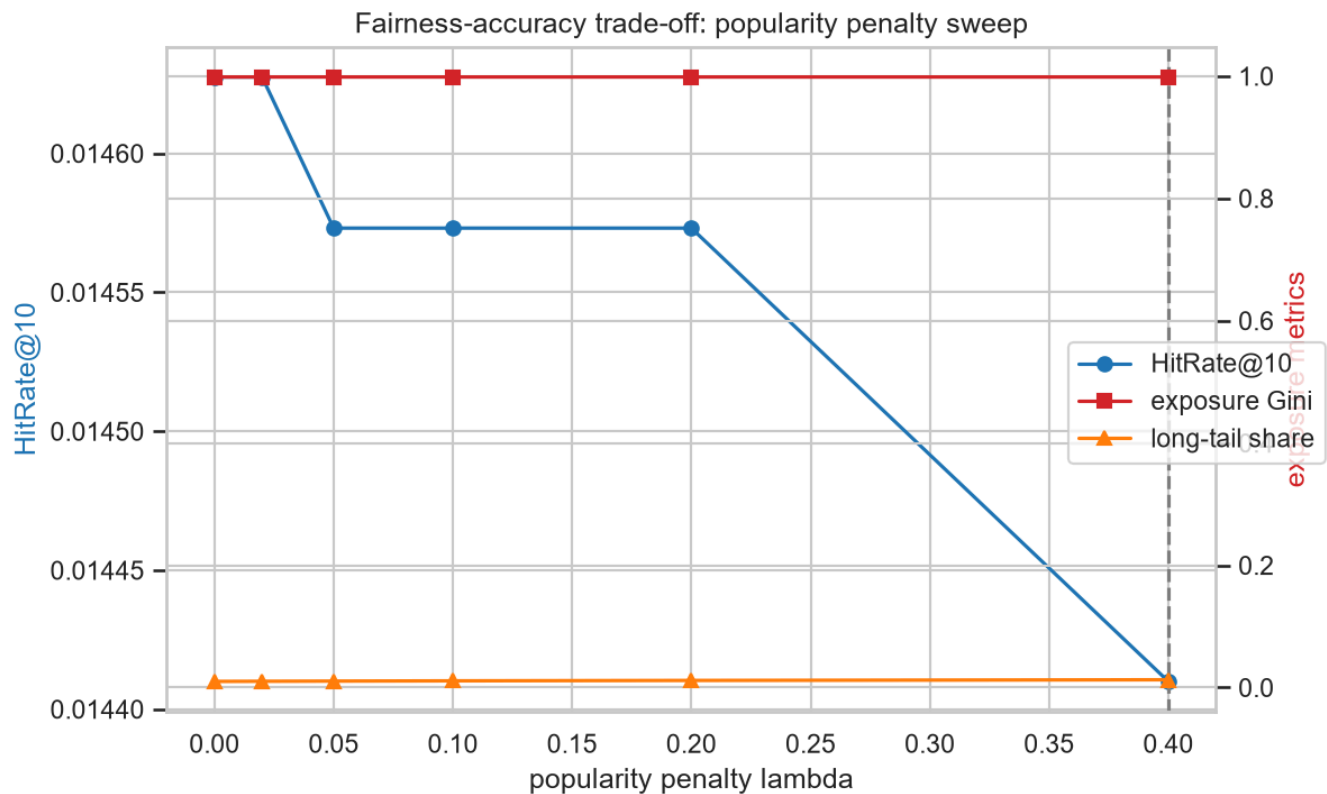
In the five-region table, TPR at  $\tau$  ranges from 0.222 [0.139, 0.319] in the Northeast down to 0.028 [0.014, 0.045] in the Southeast, a gap of 0.194, with  $n$  and bootstrap CIs in every row; North, with only 10 positives, is suppressed rather than reported as noise, and the pre-registered N+NE merge gives a gap of 0.179. Per-region AUC runs 0.825 (NE) to 0.679 (CW). Demographic parity difference (0.018) and disparate impact ratio (0.294) are reported descriptively, with the standing caveat that selection rates ride on genuinely different base rates. The finding that surprised me: the disparity *favors* the lower-income Northeast, and it persists in the final-list audit (hit@10 of 0.031 for NE customers against 0.010 to 0.014 everywhere else). The retrieval-recall column explains the mechanism. The Northeast's advantage starts at candidate generation, where regional popularity lists match NE demand best (recall 0.043 vs 0.025 in the South), before the ranker ever scores a pair. "The poor region gets better recommendations" is not the disparity direction anyone would have assumed without measuring.

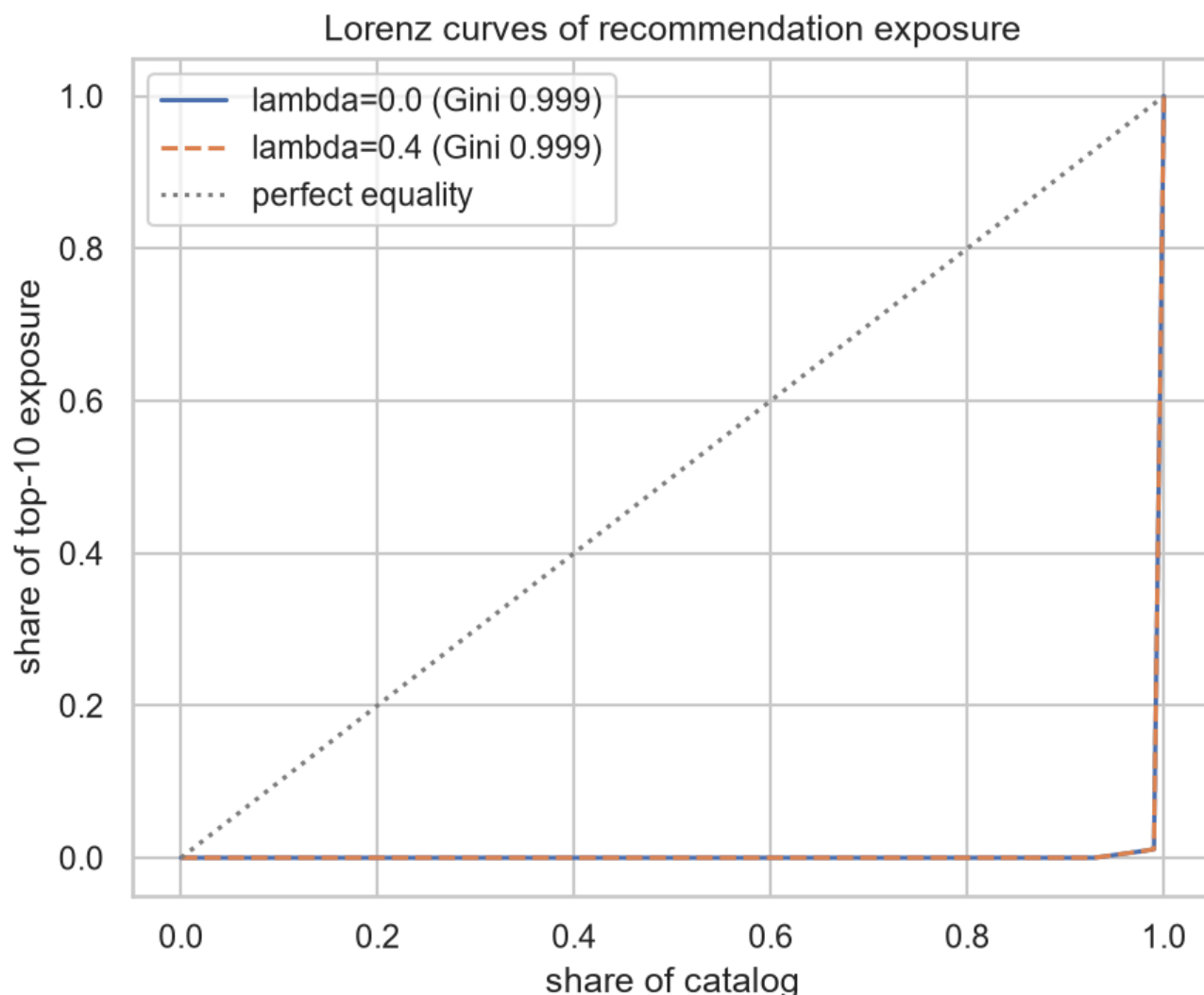
Two more checks stress-test the audit. Fairness through unawareness fails on cue: retraining the same XGBoost without the geography-derived features costs accuracy (candidate-population AUC 0.723 to 0.713). The headline TPR-gap widening (0.194 to 0.292) needs a caveat I only appreciated on review. The no-geo model's scores sit on a different scale, so TPR at the original  $\tau$  is partly a calibration artifact. The threshold-free comparison is per-region AUC, which drops for most groups (NE 0.825 to 0.802, CW 0.679 to 0.660) while the gaps between groups barely move: geography features buy accuracy without being the source of the disparity. The fairlearn ThresholdOptimizer, fitted for TPR parity, also failed to deliver: the holdout gap moved from 0.194 to 0.260 on point estimates inside the groups' wide CIs. That is a negative result with three candidate explanations, and I can't fully separate them: the fit population was the sampled-negative training task rather than candidate lists (the very mismatch this audit avoids for measurement), the holdout sits in a later window, and the per-group positive counts are tiny.

The provider side is where the numbers get dramatic. The unmitigated pipeline's top-10 lists have an exposure Gini of 0.9986, touch 7.1% of the catalog, and give small sellers (below-median volume, 5.9% of products) just 0.02% of slots. The  $\lambda$ -penalty sweep produced a null result: at every penalty strength, Gini doesn't move. The reason is structural. Cold-route candidates are the regional top-50, all head

items, so no within-list re-ranking can reach the tail. The audit also locates the bias: exposure concentration is decided at candidate generation. The matched mitigation therefore went upstream, reserving 15 of 50 cold-route candidate slots for exploration items from regional ranks 51 to 500. It backfired, costing 73% of hits, and the SHAP and PDP story from the top of this section says why: hand the ranker tail candidates and its anti-popularity gradient promotes them wholesale. The design that survives the diagnosis injects exploration *after* ranking: keep the re-ranked top-7, reserve three slots for seeded regional exploration items. Its trade-off is one a marketplace can actually price: hit@10 falls 14.5% relative (0.0146 to 0.0125) while coverage rises from 7.1% to 10.2%, long-tail share doubles, small-seller exposure more than quadruples ( $\times 4.3$ ), and Gini eases to 0.9921.

Two honesty notes on that mitigation. It was the third design tried against the same holdout, so the choice is adaptive in exactly the way I criticized for warm-cohort routing in Step 4; the selection optimism is bounded (the hit cost is structurally capped by whatever hits sat at ranks 8 to 10, and neither the slot count nor the sampling was tuned against holdout hits), but a fresh-window validation is required before anyone ships it, and it joins the future-work list on those terms. And the paired per-region deltas show the cost lands unevenly: real for the South (-0.0045 [-0.0078, -0.0021], roughly 44% of its hit rate) and the Northeast, within noise for Center-West and North. A provider-side fix that quietly taxes southern customers most is exactly the kind of side effect a final-list audit exists to catch.





Also for the future-work list: per-group serving thresholds validated on a fresh window, richer exploration policies (bandit-style, not uniform sampling), and re-checking the region gap once more holdout data accumulates, since most of the group cells here run on double-digit positive counts. Everything above, tables, CIs, mitigation deltas, is persisted in `models/fairness_metrics.json`.

## Step 8 (optional): Deployment & MLOps

The evaluated pipeline serves from a FastAPI app (`app/main.py`): `GET /recommend/{customer_unique_id}` runs the routed candidates through the trained XGBoost via the same `src/pipeline.py` code the audit notebook uses, so the API and the evaluation score identical logic. Cold-start handling comes free from the router: pass any unknown id and the response's `route` field says `global_popularity`. There is a `/health` probe for readiness, a small demo page at `/`, and five API smoke tests in CI covering routes,

ordering, validation and the fallback. Startup rebuilds artifacts from the committed data in a few seconds, which surprised me; I'd budgeted minutes, but the feature-window tables are small.

The MLOps layer is deliberately boring. The environment is pinned by `requirements.txt` plus a `python:3.11-slim` Dockerfile that bakes in data, configs and artifacts, so a container serves identically to the local run. Runs are config-driven (`configs/models.yaml`, `configs/windows.yaml`, frozen winner in `models/chosen_config.yaml`), every Step 4 training run sits in MLflow's local store, and CI has run lint plus the test suite on every push since the first commit. The deployment guide (`docs/deployment_guide.md`) writes down the monitoring plan, routing-share drift first, then score-distribution PSI, online hit-rate against the offline baselines, and monthly exposure metrics to catch the popularity feedback loop, and the versioning story: artifacts are committed with the code that made them, every git tag is a deployable snapshot, and rollback is redeploying the previous tag. The demo GIF in `docs/media/` is generated by `app/make_demo_gif.py` from live API responses, so the numbers it shows are real requests, reproducibly.

One honest scope line: the reserved-slot exploration mitigation from Step 5 is *not* enabled in the serving path, because I said it needs fresh-window validation and I meant it.

## Step 9 (optional): Use of Generative AI

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Generative AI shows up twice in this project as a feature, and throughout as a development assistant. The first feature landed back in Step 2: Claude (`claude-opus-4-8`) drafted the data dictionary from schema profiles computed off the real tables, and I verified and corrected it. The prompt, the raw draft and the final version are all committed under `docs/`, so the before/after is checkable rather than a claim.

The second is the "why you're seeing this" explanation feature (`src/llm_explain.py`). The ranker already produces per-item SHAP attributions (xgboost's `pred_contribs` is TreeSHAP, the same attribution method as the Step 5 audit), so each recommendation comes with its top five signals for free. I translate those into plain English ("customers with similar purchases also bought this product") and hand Claude the product context plus the signals under strict rules: one sentence, grounded only in what's provided, no jargon, nothing invented. The blurbs serve through the same `/recommend` endpoint behind an `explain=true` flag for the top three items, and every response is cached in `models/explanations_cache.json`, which is committed. That keeps cost and latency trivial (the committed cache holds just nine blurbs, and that covers everything the demo needs) and means the Docker container serves explanations offline; an LLM outage returns `explanation: null` instead of failing the request. No customer identifiers ever go to the API, only aggregated signals.

What went wrong is the part worth reading. My first version glossed each feature by name and ignored its value, and the very first cold-start blurb claimed the product was "popular with shoppers in your Northeast region" for a visitor whose region the system doesn't even know (the bad output is preserved



verbatim at the top of `docs/llm/explanation_examples.md`). The SHAP signal was real, `region_Northeast` was a top attribution, but its value was 0. The model was reacting to the shopper *not* being from the Northeast, and my gloss asserted the opposite. Fixing it properly took two passes. Value-aware glosses came first, so an absent binary signal gets negated text like "the shopper is not from the Northeast region". Then I noticed the negation still overclaims for unknown visitors: their region one-hots are all zero because nothing is known about them, so "not from X" is itself a claim I can't back. Those now collapse to a single "region is unknown" line, tests pin both behaviors, and the shipped cold-start blurbs make no region claims at all.

Two honest residuals survive in the committed examples. The LLM still drifts softly: one blurb turns 86 purchases plus a 4.21 average review into "solid reviews from dozens of shoppers", which quietly conflates buyers with reviewers. And it reframes negative signals as selling points. The committed prompt example shows `category_match` actually *pulled the score down* for the top warm-customer item (notebook 04 finds the same negative gradient model-wide), yet the blurb offers the category as a reason to buy. My prompt rules permit honest reframing, so this is within spec, but it shows the gap between faithful attribution and persuasive copy. Grounding rules stop invented facts; they don't stop spin. If this shipped for real I'd want an automated verification pass over each blurb, not just prompt rules.

Beyond the features, Claude assisted development itself: code scaffolding, review passes and documentation drafting, as disclosed in the README. Every analysis decision, number and conclusion in this report comes from code I ran and checked. The combined demo video (`docs/media/demo.mp4`, generated by `app/make_demo_video.py` from live in-process API responses) walks through the three recommendation routes and the explain flow end to end.

## Closing note

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Looking back at the rubric, the work I'd point a reader to isn't any single model. It is the decision discipline. Every claim that mattered got a paired per-customer difference with a bootstrap CI, and the project keeps its negative results in view: threshold optimization didn't transfer to holdout, fairness-through-unawareness made nothing better, and my first bias mitigation backfired badly enough (-73% of hits) that diagnosing *why* became the most interesting analysis in Step 5. A capstone that only reported what worked would have been thinner than this one.

Beyond the core steps, both optional tracks shipped end to end. The evaluated pipeline serves from a container with monitoring and rollback written down, and the Step 9 explanations turn the same SHAP attributions the audit uses into cached, grounded plain-language blurbs, with the LLM's failure modes documented rather than hidden. The seller-exposure trade-off is quantified to the point where it is a business decision with numbers attached: coverage 0.071 to 0.102, small sellers  $\times 4.3$ , hit rate -14.5%, and it states who pays the cost. Everything regenerates from seeded code and committed configs.

If I had another month I'd spend it on the two loose ends I flagged: validating the reserved-slot mitigation on a fresh window, and running the route-conditional re-rank experiment for warm customers. Both are one-notebook jobs on top of what is here.